

JHANVI PATEL

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EDUCATION

Adani University

Bachelor of Technology in Computer Science and Engineering | CGPA: **8.89/10**

Gujarat, India

2024 – 2028

TECHNICAL SKILLS

Languages: Go, Java, Python, JavaScript, SQL, C

Competitive Programming: LeetCode: **700+** solved, rating **1816**, **top 6.93%** | Codeforces: **Specialist**, max rating **1500**

Distributed Systems: CRDT, Redis Pub/Sub, WebSockets, event sourcing, CQRS, horizontal scaling

Concurrency & Systems: Java NIO, goroutines, channels, mutex, atomic ops, circuit breakers, rate limiting

Databases & Storage: PostgreSQL, MongoDB, SQLite, Redis, JDBC, connection pooling

Tools & Infrastructure: Git, Docker, Maven, GitHub Actions CI/CD

CS Fundamentals: Data structures & algorithms, OS, computer networks, OOP

AI & ML: XGBoost, Random Forest, Linear Regression, scikit-learn, NumPy, Pandas

PROJECTS

DocStream — Go · Redis Pub/Sub · WebSockets · PostgreSQL · CRDT · Next.js

[GitHub](#) ↗ [Live](#) ↗

- Built a distributed real-time collaborative editor in Go using a custom **RGA Sequence CRDT** with lexicographic tiebreaking, ensuring convergence across concurrent writers without a central coordinator.
- Designed horizontal scaling via Redis Pub/Sub with instance-UUID echo-loop prevention; synchronized CRDT state across multiple backend nodes with append-only PostgreSQL ops log and periodic snapshotting to minimize cold-start replay cost.
- Engineered token bucket rate limiting, CORS guards, graceful SIGTERM draining, and OpenTelemetry instrumentation across a chi-routed REST and WebSocket API.

GitResolve — Go · go-tree-sitter · SQLite · CI/CD

[GitHub](#) ↗ [Docs](#) ↗

- Built a Git merge-conflict resolution CLI using **go-tree-sitter AST parsing** to classify conflicts into 7 deterministic categories across Go, JS, and TS with specialized per-type resolution strategies.
- Enforced safety-first writes via **os.Root** sandboxing (CWE-22), **flock(2)** advisory locking, atomic file swaps, and pre-write syntax validation with a hard 10MB DoS gate.
- Persisted every conflict decision to SQLite with namespaced reason codes; CI-gateable escalation-rate metrics via **--enforce-gates**; fuzz testing, idempotency tests, and concurrent lock contention validation.

NioFlow — Java 17 · NIO · Multi-threading · Maven Central

[GitHub](#) ↗ [Docs](#) ↗

- Built an HTTP/1.1 framework from scratch using **Java NIO selector loops** and a bounded worker pool with dedicated DB executor isolation, preventing request worker starvation under concurrent JDBC load.
- Achieved **501 RPS** with **99.84% success rate** under 100 VU k6 load test (p50: 1.52ms, p99: 74.62ms); enforced **83.44% instruction coverage** via JaCoCo gate across 271 tests.
- Implemented custom circuit breakers (CLOSED/OPEN/HALF_OPEN), per-route hedging, distributed Redis rate limiting, request replay, OWASP dependency scanning, and TLS entry point with graceful **drainAndStop**.

Arbiter — Python · Database Systems · XGBoost · FastAPI · Next.js

[GitHub](#) ↗ [Live](#) ↗

- Built an ML-assisted database query optimizer that extracts **14 structural and EXPLAIN QUERY PLAN features** to predict query latency and compare alternative execution strategies.
- Evaluated Linear Regression, Random Forest, and XGBoost models using **template-split validation** to prevent query-template leakage and measure generalization on unseen SQL workloads.
- Implemented cost-based rewrite evaluation including temporary index simulation via transaction rollback, subquery-to-JOIN transformations, and per-user isolated SQLite workspaces with JWT authentication and MongoDB telemetry logging.

EXPERIENCE & LEADERSHIP

DSA Domain Lead — ASPDC, Adani University

Aug 2025 – Present

- Taught Sliding Window, Prefix Sum with HashMap, and Greedy patterns to peers; conducted workshops on pattern recognition.

Robotics Trainer — AURC, Adani University

Aug 2025 – Present

- Taught bitmask techniques for state representation and combinatorial task optimization with practical hardware and software examples.